MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL, NAND GATES, MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/1E, dated 1 June 1982, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* Heading: Under SUPERSEDING delete "MIL-M-0038510/10(USAF)" and substitute "MIL-M-0038510/1D(USAF)".

1.1, second sentence, delete and substitute: "Two product assurance classes and a choice of case outlines and lead finishes are provided for each type and are reflected in the complete part number."

1.3, Thermal resistance, delete and substitute:

"Thermal resistance, junction-to-case (θ_{JC}):
Cases (any case in M38510)- - - - - - - (See MIL-M-38510, appendix C)
Cases (all LCC's) - - - - - - - - - - - 0.08°C/W 2/"

1.3, after "Junction temperature (T_J): Add "3/".

At bottom of page, add two footnotes as follows:

"2/ When a thermal resistance value is included in MIL-M-38510, appendix C, it shall supersede the value stated herein.

"3/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions per method 5004 of MIL-STD-883."

PAGE 3

TABLE I, Conditions heading: Add "$-55°C \leq T_C \leq +125°C$".

PAGE 4

TABLE II: Delete Class C devices column in its entirety.

4.2c, delete and substitute: "c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510."

PAGE 5

4.4.3c(1): Add "or E" as deleted in Amendment 1.

6.1: Delete in its entirety.

PAGE 12

TABLE III, heading: Delete "(pins not designated are open)" and substitute "(high $\geq$2.0 V, low $\leq$0.8 V or open)." Make this same change to pages 13 through 29.

PAGE 23

* TABLE III, test numbers 39 through 46, 3B column: Delete "GND" and substitute "5.5 V".

PAGE 27

* TABLE III, test numbers 37 and 38, GND column: Add "GND" (2 places).

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:44Z
Check the source to verify that this is the current version before use.

6.4, GND: Delete "Electrical ground (common terminal)" and substitute "Ground zero voltage potential".

The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - SH, OS, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - CG, MC, AS

Agent:
DLA - ES

Preparing activity:
Air Force - 17

(Project 5962-0828)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:44Z
Check the source to verify that this is the current version before use.